Cardiorespiratory Physiotherapy — OSCE Preparation

Cases G & H | Target 70%+ (Very Good) | All reasoning uses "I ask / do / choose this because…" language

Case G — Mr. Arthur Chen | 69M | Idiopathic Pulmonary Fibrosis (IPF)

Retired stonemason. Dry non-productive cough, progressive breathlessness now limiting dressing. Digital clubbing. Fine velcro crackles both lung bases. Restrictive spirometry, reduced lung volumes. Task: assess functional capacity; provide breathlessness management and energy conservation.

1 — SUBJECTIVE ASSESSMENT

CONSENT / OPENING STATEMENT

> *"Hello Mr. Chen, my name is [Name] and I'm a physiotherapist. Today I'd like to ask you some questions about your breathing and how it affects your daily life, and then we'll look at some strategies to help manage your breathlessness. Everything is confidential. Is that okay with you? Do you have any questions before we start?"*

KEY QUESTIONS (5–6) + CLINICAL REASONING

1. "Can you describe your breathlessness — when did it first start, and has it been getting gradually worse over months or years?"

I ask this because IPF is defined by insidious, progressive onset over years — unlike acute-onset pathology (e.g., pulmonary oedema). Confirming slow progression supports the fibrotic diagnosis and helps me understand current disease trajectory.

2. "How far can you walk on the flat before you need to stop, and how does that compare to six months ago?"

I ask this because it establishes baseline functional exercise capacity and rate of decline — both critical for setting realistic rehab goals and monitoring IPF progression. It also gives me a proxy 6MWT estimate before formal testing.

3. "Which daily tasks cause you the most breathlessness — for example, dressing, showering, climbing stairs?"

I ask this because identifying specific ADL limitations allows me to prioritise energy conservation strategies and set meaningful, patient-centred functional goals. Dressing is

already flagged, so I want to map all affected activities.

4. "During your time as a stonemason, were you regularly exposed to stone dust, silica, or asbestos — and did you wear respiratory protection?"

I ask this because occupational silica dust is a known driver of pneumoconiosis and interstitial lung disease, which can mimic or co-exist with IPF. This is also relevant for medico-legal documentation and may influence the diagnosis if exposure was substantial.

5. "What medications are you currently taking — particularly any tablets specifically for your lungs, such as pirfenidone or nintedanib?"

I ask this because antifibrotic agents indicate confirmed IPF and slow progression. They also cause side effects (GI symptoms, photosensitivity) that may affect exercise tolerance and adherence to a rehab programme.

6. "Do you have a home pulse oximeter, and do you know your usual resting oxygen level? Have you been prescribed home oxygen?"

I ask this because SpO2 at rest and on exertion is the key safety parameter for exercise prescription in IPF. Significant exertional desaturation (<88%) changes my plan — I may need O2 support during exercise and must avoid unsafe exertion levels.

RED FLAGS (3–5)

□ SpO2 <88% at rest or during minimal activity

Significant hypoxaemia. Impaired alveolar-capillary diffusion in IPF drops rapidly with exertion, risking arrhythmia and organ injury.

Action: Stop assessment immediately. Position upright, apply supplemental O2 if available. Refer urgently to medical team. Do not exercise without medical clearance and O2 provision.

□ Acute exacerbation of IPF — sudden severe worsening of dyspnoea over days to weeks without identifiable cause

Carries >50% in-hospital mortality. Distinct from gradual progression; can be triggered by infection, aspiration, or occur idiopathically.

Action: Do not treat in outpatient setting. Send to emergency department immediately. Inform respiratory physician urgently.

□ Haemoptysis or new-onset chest pain

IPF patients have significantly elevated lung cancer risk. Haemoptysis may also indicate pulmonary hypertension complication or PE — all common IPF co-morbidities.

Action: Do not attribute to coughing alone. Refer urgently to GP/respiratory physician. Defer physiotherapy until cause is identified.

□ Bilateral pitting oedema + elevated JVP (cor pulmonale)

Right heart failure secondary to pulmonary hypertension from chronic hypoxia. Significantly worsens prognosis and alters the safe exercise prescription threshold.

Action: Refer to GP/cardiologist. Defer exercise prescription until medically optimised (diuretics, O2). Do not exercise without cardiovascular clearance.

FVC decline >10% over 6 months (patient-reported worsening lung function tests)

Indicates rapid disease progression — strongly predicts increased mortality in IPF. Changes the physiotherapy focus from rehabilitation to palliative breathlessness management.

Action: Flag to referring respiratory physician. Adjust goals towards symptom management, quality of life, and advance care planning rather than functional improvement.

PROBLEM LIST

PHYSICAL IMPAIRMENTS

- Reduced lung compliance (stiff fibrotic lungs)
- Impaired gas exchange — reduced DLCO
- Exercise-induced hypoxaemia / exertional desaturation
- Reduced respiratory muscle endurance
- Progressive deconditioning and reduced aerobic capacity
- Digital clubbing (marker of chronic hypoxia)

FUNCTIONAL ACTIVITY LIMITATIONS

- Breathlessness limiting dressing and personal care
- Reduced walking tolerance and community mobility
- Difficulty with stairs and inclines
- Reduced participation in leisure / social activities
- Anxiety related to breathlessness (breathlessness-fear cycle)
- Reduced quality of life

2 — PATHOPHYSIOLOGY

What is happening in the lungs: IPF is a chronic, progressive interstitial lung disease characterised by abnormal wound-healing responses in the alveolar epithelium. Repeated micro-injuries to type II pneumocytes trigger aberrant fibroblast and myofibroblast activation, leading to excessive collagen deposition and parenchymal remodelling. This fibrosis thickens and stiffens the alveolar walls, dramatically reducing lung compliance — the lungs require far greater muscular effort to inflate for the same volume of air. The thickened alveolar-capillary membrane increases the diffusion distance for oxygen, impairing gas exchange and causing hypoxaemia that worsens substantially on exertion. The restrictive spirometric pattern (reduced TLC, FVC, and DLCO with preserved or elevated FEV1/FVC ratio) directly

reflects this loss of functional parenchyma. Over time, hypoxic pulmonary vasoconstriction drives pulmonary hypertension, which can lead to right ventricular failure (cor pulmonale).

Why this patient fits: Mr. Chen's career as a stonemason provides a plausible history of dust exposure contributing to alveolar micro-injury. His fine bilateral basal velcro crackles are the pathognomonic auscultation finding of IPF — caused by the sudden snapping open of stiff, fibrotic alveoli in late inspiration. Digital clubbing reflects years of chronic hypoxia from impaired gas exchange. His progressive exertional dyspnoea, now limiting dressing, maps precisely onto the restrictive mechanics and exertional desaturation expected in moderately advanced IPF. The confirmed restrictive spirometry clinches the mechanical explanation for his symptoms.

3 — DIFFERENTIAL DIAGNOSES

Differential 1: Hypersensitivity Pneumonitis (HP)

- Also causes restrictive pattern, basal crackles, and progressive dyspnoea in older males.
- Associated with repeated exposure to inhaled organic antigens (e.g., bird droppings, mould, hay) — plausible given outdoor occupational history.
- HRCT typically shows upper-lobe predominance, ground-glass opacities, and mosaic attenuation rather than the basal-predominant honeycombing and traction bronchiectasis of IPF.
- Often has episodic acute/subacute exacerbations rather than relentless insidious progression.

> *"I favour IPF over HP because Mr. Chen's breathlessness is relentlessly progressive rather than episodic, there is no clear identified organic antigen in his history, and his demographic (69M, heavy occupational dust exposure, no bird or mould contact) strongly favours the UIP pattern consistent with IPF."*

Differential 2: Non-Specific Interstitial Pneumonia (NSIP)

- Also causes progressive dyspnoea, restrictive pattern, and bilateral crackles — clinically very similar to IPF.
- NSIP is more common in women and strongly associated with connective tissue diseases (e.g., rheumatoid arthritis, systemic sclerosis, Sjögren's).
- HRCT shows bilateral symmetrical ground-glass opacities with subpleural sparing — contrasting with the honeycombing and peripheral, basal-predominant fibrosis of IPF.
- Typically more steroid-responsive and carries a better prognosis.

> *"I favour IPF over NSIP because Mr. Chen is a 69-year-old male with no history of connective tissue disease, joint swelling, rash, or Raynaud's phenomenon. His age, sex, occupational history, and clinical presentation with velcro crackles make the UIP/IPF pattern significantly more likely on clinical grounds."*

4 — OBJECTIVE ASSESSMENT & SPECIAL TESTS

Test 1: Six-Minute Walk Test (6MWT)

- **What it measures:** Submaximal functional exercise capacity; identifies exertional desaturation; provides a prognostic marker and baseline for pulmonary rehab outcome monitoring in IPF.
- **How to perform:**
  1. Patient rests seated for 10 minutes. Record baseline SpO2, HR, BP, and Borg dyspnoea score (0–10).
  2. Mark a 30m flat corridor with cones at each end.
  3. Standardised instruction: "Walk as far as possible in 6 minutes at your own pace — you can slow down or stop if needed."
  4. Offer standardised encouragement only at each minute: "You're doing well." Monitor SpO2 continuously with pulse oximeter.
  5. Record total distance (metres), stopping points and reasons, SpO2 each minute, Borg at completion.
  6. Patient sits and rests; monitor recovery SpO2 and time to return to baseline.
- **Positive finding / clinical meaning:** Distance <400m indicates significantly impaired functional capacity in IPF. SpO2 drop ≥4% or fall below 88% on exertion = significant exertional desaturation — a key prognostic indicator associated with increased mortality. Minimal important difference (MID) = 24–30m for monitoring rehab response.
- **I select this test because:** "The 6MWT is the gold-standard functional outcome measure in IPF and pulmonary rehab. It directly informs whether supplemental O2 is needed during exercise, establishes a safe exercise prescription intensity, and the distance provides a meaningful, patient-centred goal I can track across the rehab programme."

> **Auscultation findings expected:** Fine, bilateral, late-inspiratory basal velcro crackles — the hallmark of IPF (stiff fibrotic alveoli snapping open). No wheeze. Globally reduced breath sounds reflecting restricted volumes. Post-exercise crackles may increase briefly as previously underventilated alveoli recruit.

RR: 22–28/min (tachypnoeic) | SpO2: watch for <88% on exertion | Accessory muscles: scalenes, SCM active | Shallow, rapid breathing pattern

Outcome measure: 6MWT distance in metres — repeat at 6 and 12 weeks. Also use K-BILD (King's Brief Interstitial Lung Disease questionnaire) for quality of life; Borg Dyspnoea Scale pre/post each session.

Test 2: Thoracic Expansion / Chest Excursion Measurement

- **What it measures:** Quantifies reduction in thoracic compliance and chest wall expansion — objectively confirms restrictive mechanics and provides a baseline for monitoring the impact of breathing exercises.
- **How to perform:**
  1. Patient seated upright or standing.
  2. Place a tape measure around the chest at the 4th intercostal space anteriorly (approximately nipple line).
  3. Ask the patient to exhale fully — record the circumference at maximal expiration.
  4. Ask the patient to inhale maximally — record circumference at maximal inspiration.
  5. Calculate difference (normal ≥5cm).
  6. Repeat at axilla level (upper zone) and xiphisternum (lower zone) to identify regional restriction.
- **Positive finding / clinical meaning:** In IPF, thoracic expansion is markedly reduced — often ≤2cm at the bases — reflecting fibrotic stiffness and basal volume loss. Asymmetry highlights localised restriction. Documents the physical basis of dyspnoea in a way that is tangible and communicable to the patient.
- **I select this test because:** "Chest excursion measurement gives me an objective, repeatable baseline of the mechanical restriction driving Mr. Chen's dyspnoea. It directly informs my breathing exercise prescription and lets me show him concretely — in centimetres — the impact of IPF on his chest expansion, which supports patient understanding and engagement."

> **Auscultation findings expected:** Same fine bilateral basal velcro crackles. Breath sounds globally reduced due to restricted volumes. No added sounds in upper zones — helps distinguish from COPD where wheeze would also be present.

Reduced basal expansion visually | Shallow, upper chest dominant pattern | Possible nasal flaring

Outcome measure: Tape measure chest excursion in cm at each level — documented at baseline and reassessed every 4 weeks. Compare upper vs lower zone expansion over

time.

Test 3: Five-Repetition Sit-to-Stand Test (5-STS)

- **What it measures:** Lower limb functional strength and endurance; a validated, safe proxy for exertional capacity when corridor walking is impractical; highly sensitive to exertional desaturation in ILD.
- **How to perform:**
  1. Patient seated in a standard chair (~46cm height), arms crossed over chest, feet flat on the floor.
  2. Record baseline SpO2, HR, and Borg dyspnoea score.
  3. Instruction: "Stand up fully and sit back down — do this five times as quickly as you safely can."
  4. Start timing when the patient begins to rise; stop when they return to full sitting on the 5th repetition.
  5. Monitor SpO2 continuously — stop if <85%, chest pain, or severe distress.
  6. Record time in seconds, post-test SpO2, Borg, and time to SpO2 recovery.
- **Positive finding / clinical meaning:** Time >12 seconds indicates reduced lower limb functional strength and endurance. SpO2 desaturation during STS confirms impaired cardiorespiratory reserve and is clinically equivalent to exertional desaturation on a walk test. MID in ILD for the 1-minute STS = 3 repetitions.
- **I select this test because:** "Dressing — Mr. Chen's limiting activity — involves repeated sit-to-stand equivalent movements. The 5-STS therefore has direct functional validity for his presenting complaint. It is also safer and more practical than a corridor walk when the patient is already severely limited, and it reliably detects exertional desaturation without requiring a long walking space."

> **Auscultation findings expected (post-STS):** Increased respiratory rate and amplified basal crackles immediately post-exertion. No wheeze expected. Recovery of breath sounds should be monitored — prolonged crackle persistence suggests significant ventilatory compromise.

SpO2: watch for desaturation | **RR: spikes to 28–35/min expected**

Pursed-lip breathing: spontaneous? | **Tripod posturing / forward lean post-test**

Outcome measure: 5-STS time in seconds; Borg dyspnoea score at completion. Repeat at 4-week intervals through rehab. Cross-reference with 6MWT distance for overall functional picture.

5 — TREATMENT PLAN

INITIAL PHASE — WEEKS 1–2

Airway Clearance

Not indicated as a primary technique in IPF. Mr. Chen has a dry, non-productive cough — IPF does not produce excess mucus, so ACBT or PEP would not be appropriate and may exacerbate cough distress. If cough is particularly distressing, refer to respiratory physician for pharmacological cough suppression (e.g., low-dose morphine, thalidomide in palliative stage). The physiotherapy focus should be breathlessness management only.

Breathing Exercise: Pursed-Lip Breathing (PLB) + Paced Breathing

Technique: Breathe in through the nose for 2 counts. Breathe out slowly through pursed lips (as if cooling hot soup) for 4 counts. Aim to reduce resting RR to 12–16 breaths/min. Use during all activities that provoke breathlessness.

REGRESSION Practise supine with knees supported — eliminates postural demand. Therapist hand-facilitation on lower ribs to cue lateral costal expansion. Use visual feedback (hand on abdomen vs chest) to promote diaphragmatic pattern. Begin without any activity — focus on breath awareness only.

PROGRESSION Progress to seated → standing → paced walking (2 steps inhale, 4 steps exhale). Integrate into dressing routine — breathe in between donning steps, breathe out during effort. Progress to stair-climbing rhythm (exhale on ascent, inhale at rest step).

Exercise & Mobility: Graded Walking Programme

Starting level: Begin at a pace maintaining Borg dyspnoea ≤3/10 and SpO2 ≥90% throughout. Initial target may be as short as 5–10 minutes at slow pace. Use supplemental O2 during exercise if exertional desaturation below 88% was confirmed on 6MWT.

REGRESSION If upright walking causes immediate SpO2 drop, begin with seated lower-limb exercises (seated marching, knee extensions, heel raises). Use very short bouts (2–3 min) with full rest breaks between. Titrate O2 flow rate if prescribed.

PROGRESSION Increase duration by 2–5 min per week as tolerated. Add gentle inclines. Introduce unsupported upper-limb exercises (important — avoid upper limb work that competes with diaphragm, e.g., no heavy overhead lifting). Add resistance training (elastic bands, light weights) for peripheral muscle strength once aerobic base is established.

Energy Conservation Strategies

- **The Four Ps — Plan, Prioritise, Pace, Position:** Break tasks into stages. Prioritise essential activities and delegate or simplify others. Plan rest periods before, not just after, demanding tasks.
- **Positioning for breathlessness relief:** Forward-lean sitting (elbows on knees or table) — fixes the shoulder girdle so accessory muscles assist breathing rather than posture. High side-lying with head elevated. Recovery position against a wall during acute episodes.
- **ADL modification for dressing:** Sit to dress (eliminates postural sway energy cost). Loose front-fastening clothing. Long-handled aids — sock aid, dressing stick, long-handled shoe horn — to avoid bending and breath-holding. Dress lower body first while energy is highest.
- **Home modifications:** Consider shower stool, grab rails, commode if needed, and moving bedroom downstairs as disease progresses.

Patient Education — Key Messages

- **1. Understanding IPF:** "Your lungs have scar tissue that makes them stiffer — this is why you get breathless quickly. While we cannot reverse the scarring, we can train your body to work more efficiently and help you do more with less breathlessness."
- **2. Breathlessness is not dangerous in itself:** "Feeling breathless during activity is expected with your condition — it does not mean you are harming yourself. Pacing yourself lets you stay active without triggering panic."
- **3. Staying active matters:** "Rest alone causes your muscles to weaken, which makes breathlessness worse. Gentle, regular movement within your limits is the most effective way to preserve your independence."
- **4. When to seek urgent help:** Provide a written SpO2 action plan — if resting SpO2 drops below 88% persistently, or there is sudden acute worsening of breathlessness, seek emergency review immediately. Keep a symptom diary and note any changes to report to the respiratory team.

Progression Criteria (to advance to next phase):

- SpO2 ≥90% throughout exercise at current level for two consecutive sessions
- Borg dyspnoea ≤3/10 at completion of 10-minute walk
- RR returns to baseline within 3 minutes of stopping exercise
- Patient reports confidence applying energy conservation independently

LONG-TERM PLAN — WEEKS 4–12

- **Pulmonary Rehabilitation:** Refer to a supervised PR programme (8–12 weeks, twice weekly). ILD-specific PR programmes show improved 6MWT distance, dyspnoea scores, and quality of life even without lung function change.
- **Supplemental O2:** If 6MWT confirms exertional SpO2 <88%, liaise with respiratory physician for ambulatory oxygen prescription to enable higher-intensity exercise in rehab.
- **Self-management plan:** Written home exercise programme, pacing strategies, breathlessness positions, red flags, and SpO2 action plan provided in written form.
- **Advance care planning:** As IPF is progressive, the MDT (respiratory nurse, palliative care) should initiate ACP discussions proactively — not at first presentation, but as part of ongoing care.
- **Discharge criteria:** Patient independently applying pacing and energy conservation; achieved meaningful functional goal (e.g., dresses with Borg ≤3); enrolled in pulmonary rehab; has written self-management plan and knows red flag criteria.

Examiner Demonstration — Techniques I can show in the room:

1. **Forward-lean recovery position:**

   "Mr. Chen, sit with your elbows resting on your knees and lean slightly forward — just like this. This position uses your shoulder muscles to help open your chest so breathing takes less effort. Use it whenever you feel suddenly breathless."

2. **Paced pursed-lip breathing:**

   Model the technique — inhale through the nose (count 1–2), exhale slowly through pursed lips (count 1–2–3–4). Have patient mirror. Demonstrate integrating into a walking rhythm. Correct if patient puffs cheeks or forces air — the aim is gentle, slow exhalation.

Case H — Mr. David Thompson | 55M | Chronic Bronchitis (Type B COPD — "Blue Bloater")

Bus driver. Productive cough ≥3 months/year for 2+ consecutive years. Bluish lips, bilateral peripheral oedema lower legs. High BMI. Smokes 15 cigarettes/day. Currently congested. Seeking airway clearance advice and smoking cessation support to reduce sick leave.

1 — SUBJECTIVE ASSESSMENT

CONSENT / OPENING STATEMENT

> *"Hello Mr. Thompson, my name is [Name] and I'm a physiotherapist. I understand you've been having trouble with chest congestion and it's been affecting your work. Today I'd like to ask you some questions about your breathing, show you a technique to help clear your chest, and talk about your smoking. Everything is confidential. Is that okay? Any questions before we start?"*

KEY QUESTIONS (5–6) + CLINICAL REASONING

1. "Can you describe your cough — how much phlegm do you produce per day, and what colour is it? Has the colour changed recently?"

I ask this because the clinical definition of chronic bronchitis requires a productive cough for ≥3 months per year over ≥2 consecutive years. Sputum colour is critical — clear or white is the chronic bronchitis baseline, while yellow or green with fever indicates an acute bacterial exacerbation requiring antibiotics rather than physiotherapy alone.

2. "How many cigarettes do you smoke per day and for how many years — and have you ever tried to stop before? What happened?"

I ask this because the pack-year history (packs/day × years) quantifies cumulative lung damage and COPD severity. Previous quit attempts reveal the patient's experience with withdrawal and barriers to cessation, which directly shapes my motivational interviewing approach and whether pharmacotherapy referral is appropriate.

3. "Do your ankles swell regularly — is it both sides, and does it improve overnight with leg elevation? Do you have any chest pain or palpitations?"

I ask this because bilateral peripheral oedema in a hypercapnic COPD patient suggests cor pulmonale — right heart failure secondary to pulmonary hypertension from chronic hypoxaemia. This is a significant comorbidity that limits safe exercise prescription intensity and requires urgent medical review. Concurrent chest pain would raise concern for arrhythmia or ischaemic heart disease, both common in COPD.

4. "How does your breathing affect your work as a bus driver — specifically, does breathlessness or fatigue affect your ability to drive your shifts, and how many sick days have you taken in the last 3 months?"

I ask this because establishing the occupational and functional impact is central to this consultation — reducing sick leave is Mr. Thompson's stated goal. It also raises a DVLA consideration: severe COPD with significant hypoxaemia or hypercapnia may affect fitness to hold a Group 2 commercial driver's licence, which I should be aware of and refer appropriately if concerned.

5. "How many chest infections have you had in the past 12 months requiring antibiotics, steroids, or a hospital stay?"

I ask this because exacerbation frequency is a key COPD severity and prognosis marker — ≥2 exacerbations/year is a GOLD threshold for intensifying treatment. High frequency also confirms that reducing infections through effective airway clearance and smoking cessation is a clinically meaningful and achievable target for Mr. Thompson.

6. "On a scale of 0 to 10, how ready are you to make a change to your smoking right now — and what feels like the biggest barrier for you?"

I ask this because it assesses readiness to change using the Motivational Interviewing framework and Prochaska's Stages of Change model. The answer tells me whether to plant information seeds (precontemplation), explore ambivalence (contemplation), or move into action planning (preparation stage). I do not want to push a quit plan on someone not ready — it reduces engagement and damages therapeutic alliance.

RED FLAGS (3–5)

SpO2 <88% at rest — significant hypoxaemia / possible type 2 respiratory failure

In a "blue bloater," chronic CO2 retention means the hypoxic drive may be sustaining respiratory effort. Over-oxygenation risks suppressing the hypoxic drive; under-treatment risks arrhythmia and organ damage.

Action: Refer urgently to respiratory physician. Do not administer high-flow O2 without medical guidance. Arterial blood gas may be required to assess PaCO2. Defer physiotherapy until medically reviewed.

Haemoptysis (blood in sputum)

A 55-year-old male smoking 15 cigarettes/day has substantially elevated lung cancer risk. Haemoptysis must not be attributed to coughing irritation alone without investigation.

Action: Do not proceed with treatment. Refer urgently to GP for same-day review, urgent chest X-ray, and likely 2-week-wait CT referral. Document and escalate immediately.

Acute severe breathlessness with confusion, drowsiness, or asterixis (metabolic flap)

Indicates CO2 narcosis / acute hypercapnic respiratory failure — a medical emergency in a chronic bronchitis patient with baseline CO2 retention. Asterixis is a hallmark of CO2 retention encephalopathy.

Action: Call emergency services immediately. Position upright. Apply low-flow O2 only if SpO2 <85%. Do not leave patient unattended. Document findings clearly for handover.

Rapidly worsening bilateral oedema + increased breathlessness + JVD = decompensated cor pulmonale

Acute right heart failure. Requires diuretics, supplemental O2, and possible hospital admission. Not amenable to physiotherapy alone until medically stabilised.

Action: Refer same-day to GP or emergency department. Note vital signs, oedema grade, cyanosis, and respiratory status in referral. Defer all active physiotherapy.

Green/yellow sputum + fever + worsening dyspnoea = Acute Exacerbation of COPD (AECOPD)

Requires medical management (antibiotics, systemic corticosteroids, bronchodilators). Physiotherapy airway clearance can be a helpful adjunct but is not a substitute for pharmacological treatment.

Action: Refer to GP for same-day assessment. Ensure patient has a written COPD action plan. Airway clearance physiotherapy can continue in parallel once medical management is initiated and the patient is stable.

PROBLEM LIST

PHYSICAL IMPAIRMENTS

- Excess mucus production — impaired mucociliary clearance
- Airflow obstruction (reduced FEV1/FVC on spirometry)
- Chronic hypoxaemia and possible CO2 retention
- Cor pulmonale / right heart strain (oedema, cyanosis)
- High BMI increasing work of breathing
- Active smoking perpetuating airway inflammation

FUNCTIONAL ACTIVITY LIMITATIONS

- Recurrent sick leave — primary presenting concern
- Reduced exercise tolerance and exertional dyspnoea
- Prolonged sitting at work worsening dependent oedema
- Possible DVLA fitness-to-drive consideration
- Nicotine dependence as a behavioural barrier
- Sleep disruption from nocturnal cough / possible OSA (high BMI)

2 — PATHOPHYSIOLOGY

What is happening in the airways: Chronic bronchitis results from sustained inhalation of irritants — primarily cigarette smoke — triggering a chronic inflammatory response in the large central airways. This causes hypertrophy and hyperplasia of the submucosal mucous glands, quantified by the Reid Index (ratio of gland layer thickness to total bronchial wall thickness — normal <0.4; chronic bronchitis >0.4). Goblet cell metaplasia also extends distally into smaller airways that do not normally secrete mucus. Together, these changes produce dramatically increased volumes of thick, viscid mucus that overwhelm the mucociliary escalator, leading to mucus pooling, bacterial colonisation, and recurrent lower respiratory tract infections. Chronic inflammation, airway wall oedema, and mucus plugging narrow the lumen, producing

the obstructive spirometric pattern (reduced FEV1/FVC) characteristic of COPD. Over time, chronic alveolar hypoxaemia drives hypoxic pulmonary vasoconstriction, leading to pulmonary hypertension, right ventricular hypertrophy, and eventually right heart failure — the "blue bloater" phenotype: central cyanosis from chronic hypoxaemia, and peripheral oedema from raised right-sided venous pressure, often with CO2 retention from a blunted ventilatory response.

Why this patient fits: Mr. Thompson meets the clinical definition of chronic bronchitis (productive cough ≥3 months/year for ≥2 years) driven by 15 cigarettes/day of sustained airway irritation. His bluish lips confirm central cyanosis from chronic hypoxaemia, and bilateral leg oedema confirms right heart compromise consistent with cor pulmonale — completing the classic "blue bloater" picture. His high BMI compounds the problem by increasing the work of breathing and predisposing to obstructive sleep apnoea, which worsens nocturnal hypoxaemia. Prolonged sitting as a bus driver also promotes dependent oedema formation and reduces daily physical activity, accelerating deconditioning.

3 — DIFFERENTIAL DIAGNOSES

Differential 1: Emphysema (Type A COPD — "Pink Puffer")

- Also a COPD phenotype occurring in heavy smokers with obstructive spirometry.
- Emphysema involves alveolar wall destruction and loss of elastic recoil — causing barrel chest, hyperinflation, and pursed-lip breathing to maintain positive end-expiratory pressure.
- Patients are typically thin, not cyanosed (maintain near-normal PaO2 by hyperventilating), and have minimal sputum production.
- CXR: hyperinflated lungs, flattened diaphragm, reduced vascular markings — contrasting with the vascular congestion and cardiomegaly possible in the blue bloater.

> *"I favour chronic bronchitis over emphysema because Mr. Thompson has the hallmark productive cough with large sputum volumes — emphysema is typically non-productive. He is cyanosed rather than pink, has peripheral oedema consistent with cor pulmonale, and his high BMI is the opposite of the typically thin emphysematous phenotype. The two often coexist in mixed COPD, but his presentation is dominated by the bronchitic pattern."*

Differential 2: Bronchiectasis

- Also causes chronic productive cough, large volumes of purulent sputum, and recurrent chest infections — clinically overlapping with chronic bronchitis.

- Bronchiectasis typically produces very large daily sputum volumes (often >30ml), is often position-dependent, and may show digital clubbing.
- CT chest reveals the diagnostic finding: permanently dilated, thick-walled bronchi — the "tram-track" sign on coronal view and "ring shadow" on axial view.
- Onset is usually linked to a predisposing event (post-pertussis, TB, pneumonia) or condition (cystic fibrosis, immunodeficiency, primary ciliary dyskinesia).

> *"I favour chronic bronchitis over bronchiectasis because Mr. Thompson's history is dominated by heavy tobacco exposure — the primary driver of chronic bronchitis — without a reported predisposing infection, immune deficit, or childhood respiratory history. His sputum volume appears consistent with chronic bronchitis rather than the copious, position-dependent sputum typical of bronchiectasis. However, I would recommend a CT chest to formally exclude bronchiectasis given his chronic productive cough and infection history."*

4 — OBJECTIVE ASSESSMENT & SPECIAL TESTS

Test 1: Assessment & Supervised Trial of ACBT (Active Cycle of Breathing Technique)

- **What it measures:** ACBT is the first-line, evidence-based airway clearance technique for chronic bronchitis. The supervised trial assesses the patient's ability to perform each phase, evaluates immediate sputum production and clearance, and identifies whether modifications or adjuncts (e.g., oscillating PEP) are needed.
- **How to perform (teach and assess — three phases cycling):**
  1. **Breathing Control (BC):** Relaxed, diaphragmatic breathing at normal tidal volume for 4–6 breaths. "Let your shoulders drop and breathe gently and normally." Provides rest and prevents fatigue between active phases.
  2. **Thoracic Expansion Exercises (TEE):** 3–5 slow, deep inhalations — breathe in as fully as possible, hold for 3 seconds (collateral channels open and air gets behind secretions), then gentle passive exhalation. May use therapist hand-facilitation on lower ribs to cue lateral basal expansion.
  3. **Forced Expiratory Technique (FET) — Huff:** Open-glottis forced expiration — "Take a medium breath in, then breathe out firmly and quickly with your mouth open, making a 'huff' sound." Perform 1–2 huffs at mid-to-low lung volume (moves peripheral secretions centrally), then 1 huff at higher lung volume (clears proximal airways). Follow immediately with Breathing Control.
  4. Repeat cycle: BC → TEE → FET → BC for 3–4 complete cycles or until sputum is cleared.

- **Positive finding / clinical meaning:** Audible, productive sputum clearance on huffing confirms the technique is mobilising secretions effectively. Assess sputum colour (clear/white = baseline; yellow/green = infection) and volume. Failure to clear with ACBT alone may indicate need for oscillating PEP (Acapella, Flutter valve) to break up tenacious secretions.
- **I select this test because:** "ACBT directly addresses Mr. Thompson's primary complaint — congestion — and is the evidence-based first-line technique for chronic bronchitis. Teaching and assessing it in this session provides him with an independent self-management tool he can use every morning before work to reduce mucus build-up, thereby decreasing the infection frequency that is causing his sick leave. It is the technique with the most immediate and functional relevance to his goals."

> **Auscultation findings expected:** Coarse, wet inspiratory and expiratory crackles throughout both lung fields — most prominent at the bases but not exclusively basal, distinguishing it from IPF. Expiratory polyphonic wheeze from airflow obstruction. Prolonged expiratory phase. Post-ACBT: crackles should reduce in extent and intensity — audible clearance confirms technique efficacy and provides immediate feedback to patient and examiner.

RR: 20–26/min at baseline **SpO2: may be 88–92% at rest**

Central cyanosis: lips and tongue **Bilateral lower leg oedema**

Accessory muscle use on exertion

Outcome measure: Leicester Cough Questionnaire (LCQ) — validated patient-reported cough impact measure; MCID = 1.3 points. Also track self-reported sputum diary (daily volume and colour) and number of sick days per month as a functional outcome.

Test 2: Six-Minute Walk Test (6MWT)

- **What it measures:** Submaximal functional exercise capacity in COPD; identifies exertional desaturation and dyspnoea severity; provides objective baseline for safe exercise prescription and pulmonary rehab monitoring.
- **How to perform:**
  1. Record baseline SpO2, HR, BP, Borg dyspnoea (0–10), and Borg fatigue score.
  2. 30m flat corridor with cones. Standardised instruction: "Walk as far as you can in 6 minutes — slow down or stop as needed."
  3. Standardised encouragement every minute. Monitor SpO2 continuously with pulse oximeter. Stop if SpO2 <85%, severe chest pain, or patient requests to stop.
  4. Record total distance (metres), reasons for stopping, SpO2 each minute, time to recovery of SpO2 post-test.

- **Positive finding / clinical meaning:** Distance <350m in a 55-year-old male suggests moderate-to-severe functional limitation in COPD. SpO2 desaturation below 88% on exertion confirms need for O2 assessment. MID in COPD = 25–33m. Fitness-to-drive implications should be noted if severe desaturation occurs — relevant for a bus driver.
- **I select this test because:** "I need an objective measure of Mr. Thompson's exercise capacity to safely prescribe a walking programme as part of his smoking cessation support and COPD management. The 6MWT also tells me whether he is a candidate for supplemental O2 during exercise, gives me a baseline for pulmonary rehab referral, and provides a meaningful functional number I can use to track improvement and motivate Mr. Thompson."

> **Auscultation findings expected (during and post 6MWT):** Increased expiratory wheeze during exertion from dynamic airway collapse. Coarse crackles persist throughout. Prolonged expiratory phase. Possible reduced air entry at bases if mucus plugging is significant. Wheeze should reduce at rest post-walk.

SpO2: monitor for exertional desaturation | **Borg dyspnoea at completion** | **Pursed-lip breathing on exertion?** | **RR recovery time post-walk**

Outcome measure: 6MWT distance in metres at baseline; repeat at 6 and 12 weeks. Also use COPD Assessment Test (CAT) — validated 8-item COPD symptom questionnaire; MCID = 2 points. Use in conjunction with mMRC Dyspnoea Scale.

Test 3: Full Auscultation & Baseline Respiratory Screen

- **What it measures:** Documents baseline breath sounds, wheeze, and crackle distribution — confirms the clinical picture consistent with chronic bronchitis, identifies any consolidation or asymmetry, and provides an immediate before-and-after measure of ACBT efficacy.
- **How to perform:**
  1. Patient seated upright, arms loosely crossed in front.
  2. Use the diaphragm of the stethoscope. Compare bilaterally and symmetrically — upper, mid, and lower zones; anterior and posterior chest.
  3. Ask patient to breathe through the mouth, slightly deeper than normal.
  4. Assess: air entry (present/reduced/absent), breath sound quality, added sounds (crackles — timing, pitch, distribution; wheeze — mono/polyphonic, inspiratory/expiratory), expiratory phase duration.
  5. Note any asymmetry — asymmetric crackles or absent sounds raise concern for consolidation or pleural effusion.
  6. Document RR, breathing pattern, accessory muscle use, cyanosis (lips, tongue, periphery), and pitting oedema grade.

- **Positive finding / clinical meaning:** Bilateral coarse crackles throughout — consistent with mucus in large airways. Expiratory polyphonic wheeze from obstructive narrowing. Prolonged expiratory phase (>3 seconds). Post-ACBT auscultation confirming reduced crackles = objective proof of airway clearance success.
- **I select this test because:** "Auscultation is quick, non-invasive, and guides everything else. It confirms whether Mr. Thompson's presentation is consistent with mucus in large airways (supporting ACBT) or whether there is consolidation or effusion that would change my approach and require urgent referral. Repeating it immediately post-ACBT lets me demonstrate to Mr. Thompson — and to the examiner — that the technique has worked, which reinforces motivation for daily self-management."

> **Auscultation findings expected:** Coarse bilateral crackles, expiratory polyphonic wheeze, prolonged expiration. Reduced air entry at bases if significant mucus plugging. After ACBT: crackles reduce — patient may cough productively, confirming secretion mobilisation.

RR: 20–26/min | SpO2: 88–92% at rest | Central cyanosis visible | Bilateral pitting oedema | Expiratory wheeze audible

Outcome measure: Auscultation findings documented pre and post each treatment session. Track sputum volume and colour in daily diary. Use CAT score and mMRC Dyspnoea Scale as ongoing functional and symptom burden measures.

5 — TREATMENT PLAN

INITIAL PHASE — WEEKS 1–2

Airway Clearance: Active Cycle of Breathing Technique (ACBT)

Full technique: BC (4–6 relaxed breaths) → TEE (3–5 deep breaths with 3-second inspiratory hold) → FET/Huff (1–2 huffs at mid-volume, 1 at high volume) → BC. Repeat 3–4 cycles. Perform upright seated or in high side-lying. Aim for twice daily — morning on rising (clears overnight secretion accumulation before work) and evening.

REGRESSION If acutely unwell or very breathless, simplify to Breathing Control and gentle FET only. Use gravity-assisted positioning (right or left side-lying with foot of bed elevated 15–30° for targeted lobar drainage) to assist gravity drainage before huffing. Increase rest periods between cycles. Allow patient to perform in supported recline if upright not tolerated.

PROGRESSION Add oscillating PEP device (Acapella or Flutter valve) — high-frequency airway wall vibrations loosen tenacious secretions. Progress to full independent home programme with written instructions. Coordinate ACBT 30–60 minutes after inhaled bronchodilator use for maximum airway opening before clearance. Add saline nebuliser pre-ACBT (if prescribed) to hydrate and thin secretions.

Breathing Exercise: Pursed-Lip Breathing (PLB)

Technique: Breathe in through the nose for 2 counts. Breathe out slowly and gently through pursed lips for 4 counts. PLB maintains positive pressure in the small airways on exhalation, preventing dynamic collapse — particularly important in obstructive disease where airways are prone to collapse during forced expiration.

REGRESSION Practise seated at rest with no activity demand. Use a mirror for visual feedback. Start with very slow, unrushed practice — 5 minutes at rest, twice daily. Ensure patient understands this is not a forced or strained exhalation — gentle and controlled.

PROGRESSION Integrate into walking (2 steps inhale, 4 steps exhale). Teach as a recovery strategy after activities causing dyspnoea. Progress to using during activities of daily living relevant to work — e.g., walking to the bus depot, boarding vehicles. Add diaphragmatic retraining in parallel.

Exercise & Mobility: Graded Walking Programme

Starting level: Begin at a pace maintaining Borg ≤4/10 and SpO2 ≥88%. For a high-BMI patient with likely moderate COPD, target 10–15 minutes of continuous walking at a comfortable pace. Time the session to follow morning ACBT and inhaler use for optimal airway patency.

REGRESSION If unable to maintain SpO2 ≥88% on continuous walking, use interval training — walk 3 minutes, rest 2 minutes, repeat ×3–4. Refer for ambulatory O2 assessment if consistently desaturating below 88%. Seated lower-limb exercises as an alternative aerobic stimulus (seated cycling, marching) until upright tolerance improves.

PROGRESSION Increase duration by 5 minutes every 1–2 weeks as tolerated. Aim for 30 minutes most days. Add gentle inclines. Introduce structured resistance training (elastic bands, light weights) for peripheral muscle strengthening — improves exercise capacity independently of lung function. Interval training may be better tolerated than continuous walking in moderate-severe COPD.

Smoking Cessation — Motivational Interviewing Approach

- **Assess readiness:** "On a scale of 0 to 10, how important is it to you to quit smoking, and how confident are you that you could if you decided to?" Explore what would shift the number upward — identify ambivalence, not resistance.
- **Match stage of change:**
  - *Precontemplation:* Provide non-judgmental factual information — "Smoking is the single most important factor driving your congestion and infections." Plant seeds. Do not lecture.
  - *Contemplation:* Explore pros and cons. "What would be different for you at work if you weren't getting chest infections as often?" Develop discrepancy between current behaviour and personal goals.
  - *Preparation/Action:* Set a quit date. Refer to NHS Stop Smoking Service. Discuss NRT options (patch + fast-acting gum or inhalator — combination NRT is more effective than single agent). Consider pharmacotherapy referral (varenicline — most effective cessation aid; GP prescription needed).
- **Occupational framing:** "Your job as a bus driver depends on your health certificate. Quitting is the most powerful thing you can do to slow lung damage, reduce infections, and protect your income."
- **Written resource:** Provide NHS Smokefree leaflet and local Stop Smoking Service contact details.

Patient Education — Key Messages

- **1. Why your chest gets congested:** "Your airways are producing too much thick mucus because of years of smoke irritation. Using the huffing technique I've shown you every morning — especially before work — shifts this mucus before it builds up and turns into an infection."
- **2. How to spot an infection early — and act on it:** "If your phlegm turns green or yellow and you feel worse, don't wait. Start your rescue antibiotics and steroids if you have them, and contact your GP the same day. Early treatment means shorter illness and fewer sick days."
- **3. Smoking is the priority:** "Inhalers, physiotherapy, and everything else helps manage your COPD — but quitting smoking is the only intervention that actually slows the damage. Every cigarette adds more irritation to your airways and contributes to your infections and your sick leave."
- **4. Staying active helps — even short walks:** "Regular gentle walking strengthens your breathing muscles and helps reduce the swelling in your legs. You don't need a gym — 10 minutes of walking after your morning chest clearance routine is a great start, and we'll build from there."

Progression Criteria (to advance to next phase):

- Patient performing ACBT independently and correctly twice daily at home
- Sputum diary showing consistent daily clearance; no new change in sputum colour
- SpO2 ≥88% maintained during 10-minute walk (or ambulatory O2 plan in place)
- Borg dyspnoea ≤4/10 after 15-minute walk
- Patient engaged with smoking cessation (referral accepted or NRT commenced)

LONG-TERM PLAN — WEEKS 4–12

- **Pulmonary Rehabilitation:** Refer to supervised PR programme (8–12 weeks, twice weekly). Strong evidence in COPD — PR improves 6MWT distance (+50–80m), dyspnoea, quality of life, and significantly reduces hospital admissions. The highest-impact non-pharmacological intervention available.
- **Oscillating PEP device:** Prescribe Acapella or Flutter valve for home use if ACBT alone is insufficient for tenacious secretions. Review at 4 weeks for tolerance and efficacy.
- **MDT referrals:** Dietitian (weight management — reduces work of breathing and improves functional capacity); Respiratory Nurse Specialist (COPD action plan, inhaler technique check); GP (O2 assessment, pharmacotherapy review, occupational health referral regarding commercial driver fitness).
- **Self-management plan:** Written COPD action plan — includes ACBT home programme, sputum diary, what to do during exacerbations, smoking cessation progress, exercise diary, and emergency contacts. Register with NHS Breathe Easy support group for peer support.
- **Discharge criteria:** Patient independently performing ACBT; enrolled in or completing pulmonary rehab; engaged with smoking cessation programme; 6MWT improved ≥25m or functionally stable; sick leave reduced; written COPD action plan in place and understood.

Examiner Demonstration — Techniques I can show in the room:

1. **ACBT Huff (FET) — demonstrating open vs closed glottis:**

   "Mr. Thompson, a cough slams your airways shut and can actually trap mucus — a huff keeps your mouth and throat open so mucus can move. Watch me: I take a medium breath in, then breathe out firmly with my mouth open — 'huff.' Now you try." Correct any mistakes — closed glottis (throat tightening), puffed cheeks, or forced exhalation causing wheeze. Show both mid-volume huff (clearing peripheral mucus) and high-volume huff (clearing upper airways).

2. **Pursed-lip breathing with paced walking:**

   Walk alongside Mr. Thompson — inhale for 2 steps, exhale through pursed lips for 4 steps. Cue the rhythm verbally: "In… 1, 2… out… 1, 2, 3, 4." "This stops the

breathlessness panic — it gives you back control over your breathing as soon as you feel short of breath."